1. Introduction

Objective:
Implementation of **longitudinal control** of an autonomous agent that manages traffic lights.

Human like behaviors:

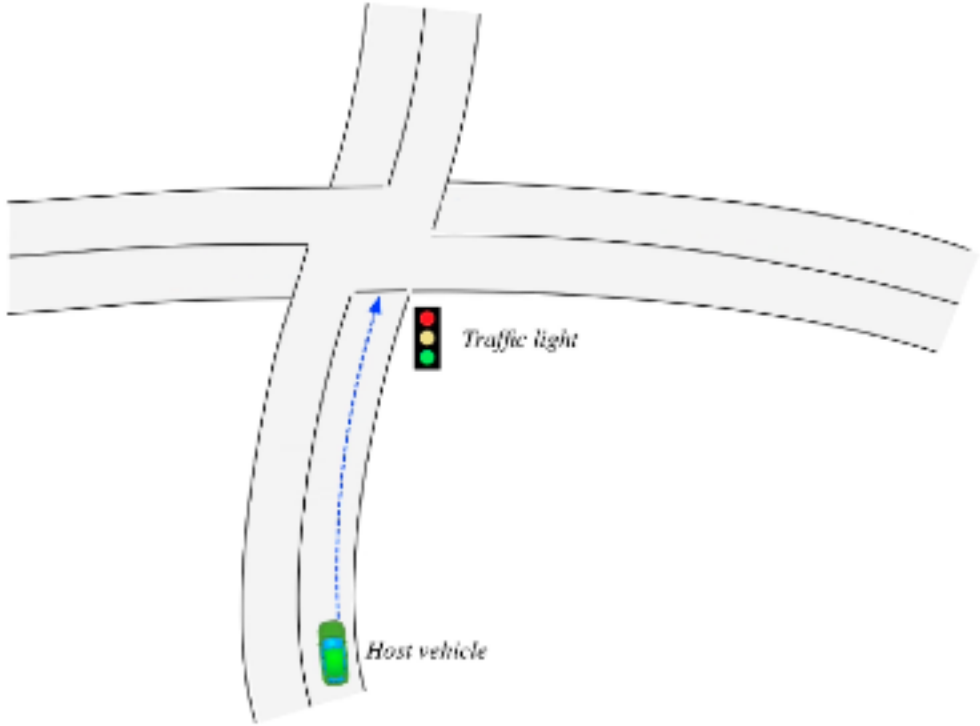
- action priming → **minimal** intervention principle

  (minimum square jerk criterion)

$$\begin{cases} \dot{s}(t) = v(t) \\ \dot{v}(t) = a(t) \\ \dot{a}(t) = j(t) \end{cases} \qquad J = \int_0^T j(t)^2 \, dt$$

$$s(t) = c_1 t + \frac{1}{2} c_2 t^2 + \frac{1}{6} c_3 t^3 + \frac{1}{24} c_4 t^4 + \frac{1}{120} c_5 t^5 \quad ;$$

- action selection → basal ganglia structure.



2. Minimum Jerk

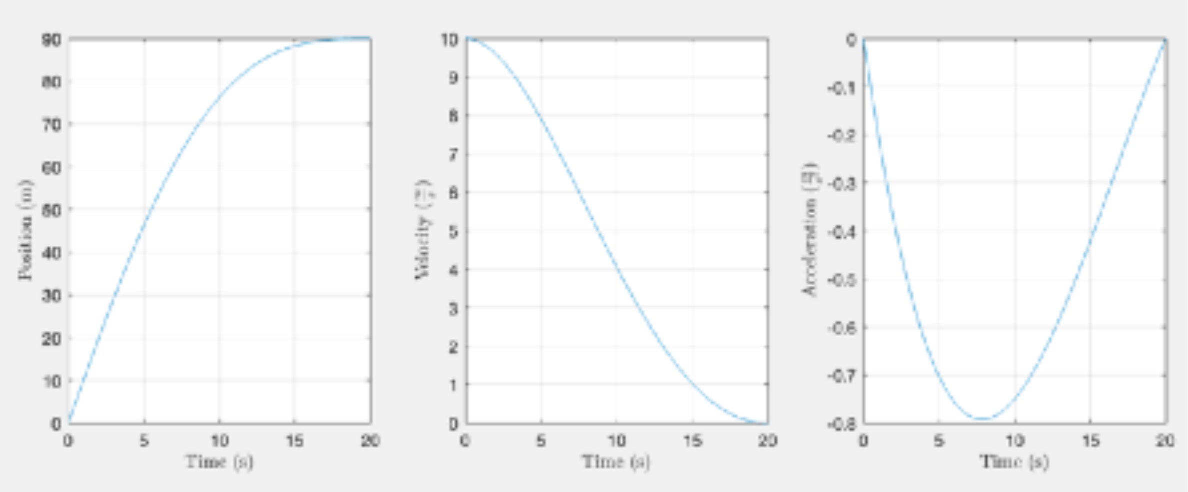
Optimal Control Problem:
Square jerk function is the Lagrangian cost of our OCP → solved using *MATLAB* imposing *initial* and *final conditions* → obtaining **optimal solutions** and extracting **opt**. **coefficients**.

$$s_{\text{opt}}(t) = c_1 t + \frac{1}{2}c_2 t^2 + \frac{1}{6}c_3 t^3 + \frac{1}{24}c_4 t^4 + \frac{1}{120}c_5 t^5$$

$$v_{\text{opt}}(t) = c_1 + c_2 t + \frac{1}{2}c_3 t^2 + \frac{1}{6}c_4 t^3 + \frac{1}{24}c_5 t^4$$

$$a_{\text{opt}}(t) = c_2 + c_3 t + \frac{1}{2}c_4 t^2 + \frac{1}{6}c_5 t^3$$

$$j_{\text{opt}}(t) = c_3 + c_4 t + \frac{1}{2}c_5 t^2$$



Subsequently, the application *MATLAB coder* has been used to generate the **C** code needed to implement the **agent**.